Analyzing and Interpreting Motion Graphs: Lab

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Minds On

- Time on x-axis, position on y-axis; right is positive direction
- Watch the walker -> sketch a p-t each round
- “Boards up” on my cue; whisper-check first

Minds-ON

Round 1: Stand at 2 m. Stay still for 5 s

Round 2: Start at 1 m. Walk steadily away for 5 s

Round 3: Start at 6 m. Walk steadily toward 0 for 5 s

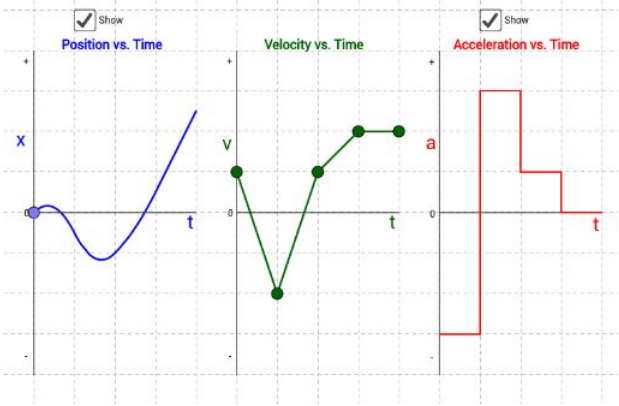
Learning Goal

- ➔ We are learning how to read and draw p-t, v-t, and a-t graphs for 1-D motion
- ➔ We are learning how to convert between these graphs using slope (for velocity and acceleration) and area (for displacement)

Success Criteria

- ➔ I can read p–t, v–t, and a–t graphs and describe the motion with correct units and sign.
- ➔ I can draw the matching graphs (p–t ↔ v–t ↔ a–t) using slope to link them.
- ➔ I can find displacement from a v–t graph (area) and cross-check it on my p–t graph.
- ➔ I can collaborate with my group by doing my role, listening, and helping my group stay on task.

Mini-Review



Big 3:

- p-t slope = velocity
- v-t slope = acceleration
- Area under v-t = displacement

Investigation 1: Toy Car Data collection and Analysis

Objectives

- Collect position-time data for a toy car
- Plot position-time graph and draw line of best-fit
- Calculate velocity from the slope of the line

Materials

- Toy car
- Stopwatch (phones, school laptops or tablets)
- Meter sticks (or measuring device of some sort)
- Masking tape
- Data recording sheet (sheet of paper for recording data)

Data Table and Graph Setup

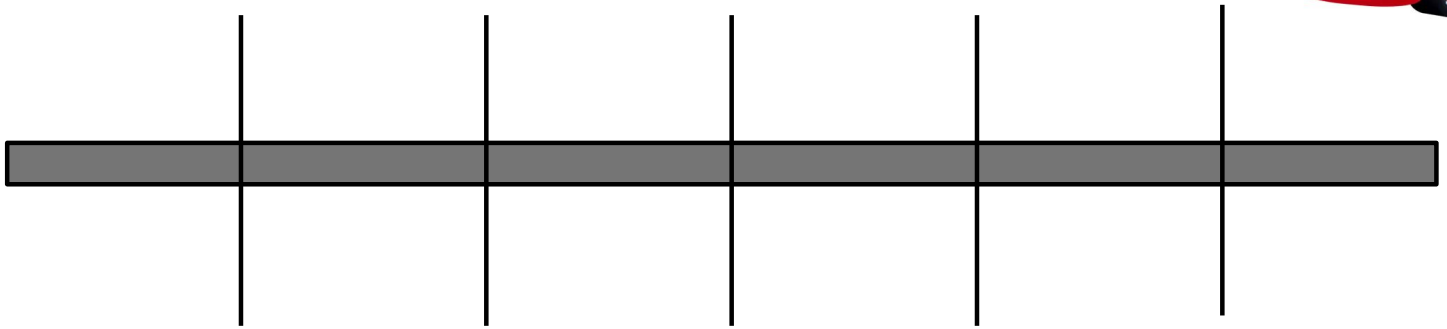
Time (s)	Position (cm)
0	0
2	40
…	…

Graph tips

- Time on horizontal (x) axis, position on vertical (y) axis
- Label axes with units
- Use ruler for straight axes and a colored pen for clarity
- Try to keep the time intervals constant (e.g., every 2 s)

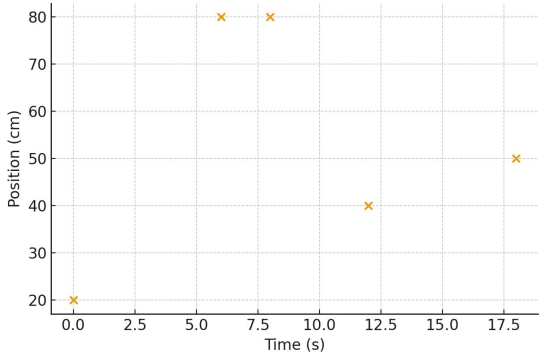
Track Setup and collecting data

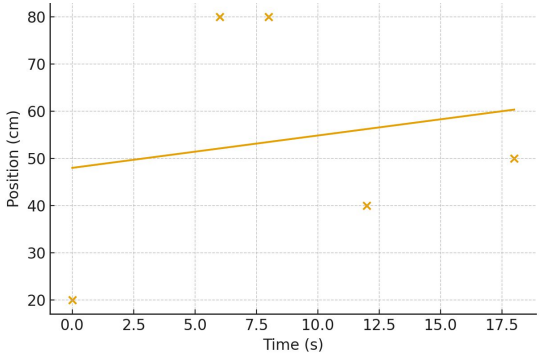
“**20 cm** mark at **2 seconds**
30cm mark at **4 seconds**”



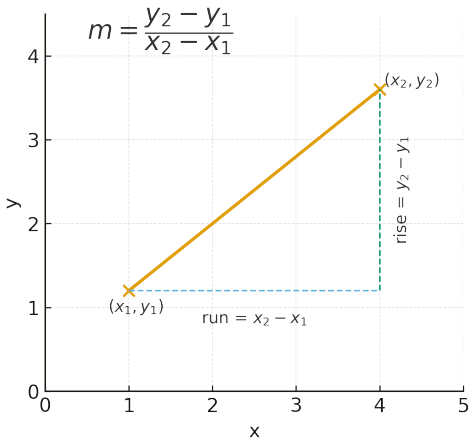
Graph Setup: Example

Time (s)	Position (cm)
0	20
6	80
8	80
12	40
18	50
…	…





Best-Fit Line: Trend Vs Noise



Calculating slope:

1. Pick any two points
2. Plug into formula
3. How can we interpret the slope? What does the slope represent?

Notes

- Do not connect dots
- One straight line that best represents the trend

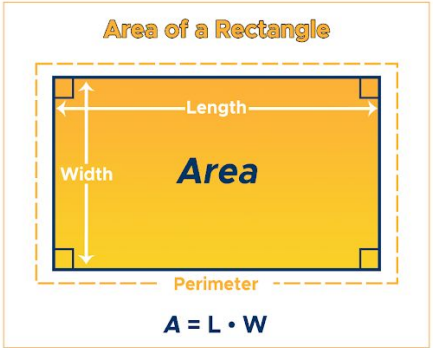
Converting p–t to v–t

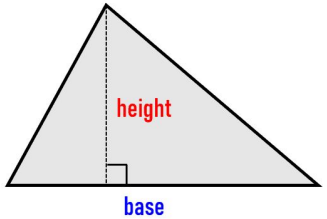
To keep in mind:

- p–t straight line with positive slope → v–t horizontal line above zero
- p–t horizontal line → v–t at zero
- p–t curve getting steeper → v–t increasing

Steps:

1. Break p–t into time segments
2. Compute slope for each segment
3. Plot those slope values as horizontal segments on v–t





$$A = \frac{1}{2}bh$$

b = base

h = height

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Area Under v–t: How to Get Displacement

Displacement = area under the v-t curve

Shapes:

- Rectangle = base × height
- Triangle = ½ × base × height

Example:

- Constant 4m/s for 5s -> 20m
- Acceleration from 0 to 4 m/s over 2s -> 4m

Slope Part 2: Getting Acceleration

Acceleration equals the slope of the v-t graph ($\Delta v/\Delta t$)

To keep in mind:

– Rising v–t → positive a–t

– Flat v–t → zero a–t

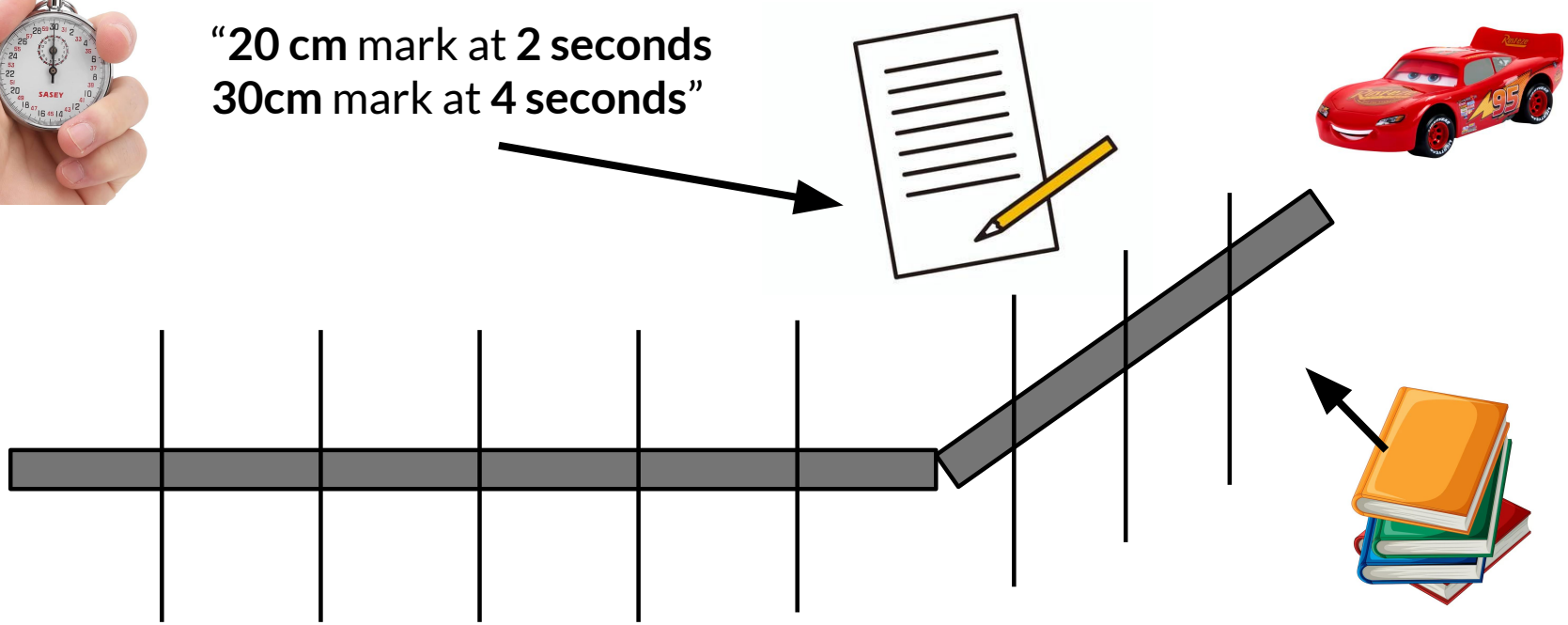
– Falling v–t → negative a–t

Steps:

1. Identify intervals on v–t.
2. Compute $a = \Delta v/\Delta t$
3. Plot those values as horizontal segments on a–t.

Track Setup and collecting data (RAMP)

"**20 cm** mark at **2 seconds**
30cm mark at **4 seconds**"



Graph Setup: Example

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0	20
6	80
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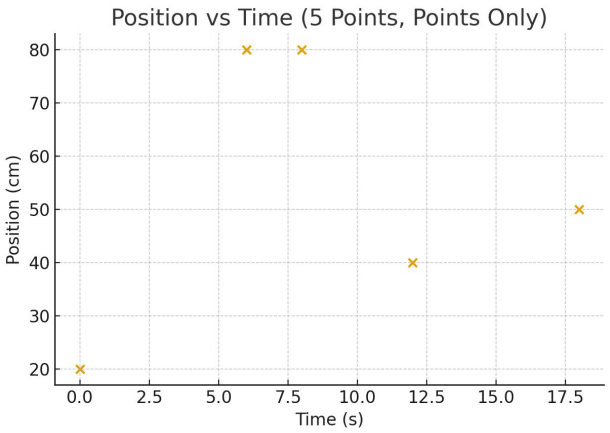
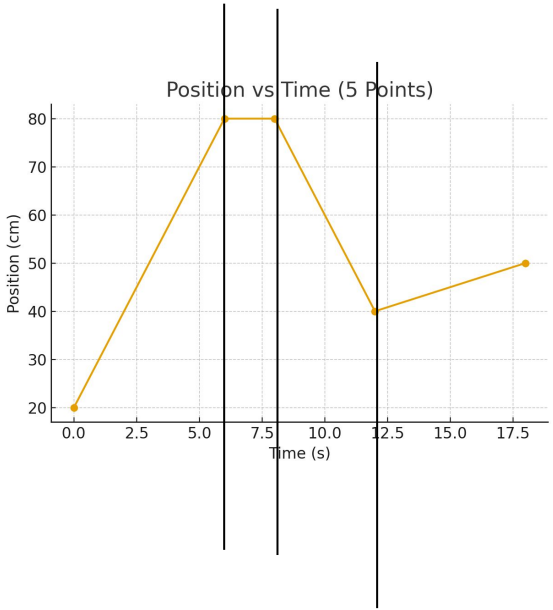
…	…



Regions



*****Note, here we will be connecting the points***

Poster Instructions (Ramp)

- Attach your three graphs (glue/tape): p–t (top), v–t (middle), a–t (bottom).
- Label regions on all three: “Region A: ramp” and “Region B: level,” with a vertical dashed transition line
- Show slopes for each region and how you calculated them
- Compute the area under your v–t over a stated time interval and state what it represents (displacement, in metres).
- Write what the slope and area represents in our motion graphs

Gallery Walk: What to Do

Visit two posters (that are not your own)

Sticky note: one accurate connection + one question you may have

Ex: Where is velocity the largest? What p-t feature matches it?

3-2-1 Connect & Reflect

3 connections using slope / area / sign (own data)

2 fixes for a partner (units, sign, axes, slope vs height)

1 question you still have

Motion Graphs Support Sheet

Supplementary support resource to use alongside the original booklet and lesson materials.

This support sheet keeps the learning goal the same for everyone. It gives students a clear task sequence, key graph reminders, helpful vocabulary, and sentence starters so they can focus on the physics thinking in the lesson.

What are we learning?

We are learning how position-time (p-t), velocity-time (v-t), and acceleration-time (a-t) graphs connect in one-dimensional motion. We are also learning how to use slope and area to explain what is happening in the motion.

Big 3 reminders

- Slope of a p-t graph = velocity
- Slope of a v-t graph = acceleration
- Area under a v-t graph = displacement

Before you start

☐ Toy car
☐ Track with visible position labels
☐ iPad or phone for slow-motion recording
☐ Student booklet
☐ Graph paper
☐ Ruler or straightedge
☐ Pencil or pen
☐ Stopwatch if needed

Group roles

Role	Main responsibility
Videographer	Records the run in slow motion, keeps the device steady, and helps the group pause at the correct frame.
Timer/Launcher	Starts the run, helps call or confirm time points, and checks that the trial begins from the right

Recorder/Grapher	place. Fills in the table, helps organize the graphing work, and checks labels, units, and graph quality.

All group members should help explain the graphs, check the final work, and rotate support where needed.

Investigation checklist

Part 1: Level track

☐ Record time and position data for Trial 1.
☐ Record time and position data for Trial 2.
☐ Graph your p-t data.
☐ Label axes and units clearly.
☐ Draw a best-fit line instead of dot-to-dot.
☐ Choose two points on the best-fit line and calculate slope.
☐ State what the slope means in the context of the trial.
☐ Draw the matching v-t graph.
☐ Draw the matching a-t graph.
☐ Explain how the graphs connect.

Part 2: Ramp then level

☐ Record time and position data for the ramp run.
☐ Mark the transition from Region A (ramp) to Region B (level).
☐ Draw the p-t graph.
☐ Describe how the shape changes from Region A to Region B.
☐ Draw the matching v-t graph.
☐ Draw the matching a-t graph.
☐ Estimate the sign or value of acceleration during the ramp.
☐ Use area under the v-t graph to estimate displacement.
☐ Cross-check that displacement with the p-t graph.
☐ Prepare your poster explanation.

Vocabulary help

Word	Helpful meaning

position	Where the object is.
velocity	How fast and in what direction position changes.
acceleration	How velocity changes over time.
slope	Rise over run; tells the rate of change.
displacement	Overall change in position.
best-fit line	One line that shows the overall trend in the data.
Region A	The ramp section of the motion.
Region B	The level section after the ramp.
positive / negative	Used to show direction based on the chosen positive direction.

Sentence starters

- The slope of our p-t graph is _____, so the velocity is _____.
- Our v-t graph is _____ because the velocity is _____.
- Our a-t graph is _____ because the slope of the v-t graph is _____.
- The area under our v-t graph from _____ to _____ represents _____.
- In Region A, the car is _____, so the graph _____.
- In Region B, the car is _____, so the graph _____.

What to check before handing in your work

☐ Are the axes labelled?

☐ Are the units included?

☐ Did we use a best-fit line where appropriate?

☐ Did we explain slope correctly?

☐ Did we explain area on the v-t graph correctly?

☐ Did we mark Region A and Region B clearly?

☐ Did we use signs correctly?

3-2-1 Writing Support

Supplementary support resource for the existing consolidation and poster explanation tasks.

Students may use this sheet to organize their thinking before they complete the original 3-2-1 handout. It supports communication without lowering the physics expectation.

How to use this page

Use your own graphs, numbers, regions, and times. You may write in full sentences or point form if your teacher has already said that is an appropriate option for you.

A. Three connections

1. On our level run, the slope of the p-t graph was _____, so the velocity was _____.
2. In Region A, the p-t graph _____, which means the v-t graph _____ and the a-t graph _____.
3. From time _____ to _____, the area under the v-t graph was _____, so the displacement was _____.

B. Two fixes or a glow and next step

- One thing that was strong in my partner's work was _____.
- One next step that would improve the work is _____.
- Another strength or next step is _____.

C. One reflection

- One strategy that helped me today was _____.
- One place I got stuck was _____, and I fixed it by _____.
- I am still wondering why _____.

Quick self-check

☐ Did I use the words slope, area, velocity, acceleration, or displacement correctly?

☐ Did I include units where needed?

☐ Did I explain the graph meaning, not just how the graph looks?

☐ Did I connect my answer to my own trial, graph, region, or time interval?

Motion Graphs Lesson: Learner Support Notes

Teacher-use resource to accompany the lesson plan and original classroom materials.

This page uses a strengths-based lens. The focus is on what helps each learner participate well, communicate understanding, and access the same physics expectation as the rest of the class.

Whole-class reminder

All students are working toward the same learning goal and success criteria. The supports in this lesson are intended to improve access, communication, and participation without lowering the expectation. The focus stays on graph interpretation, slope, area, units, sign, and communication of reasoning.

Student-specific notes

Student	Strengths-based view	Helpful supports in this lesson
Michael	Michael is social, friendly, hardworking, and a strong self-advocate. He is likely to show his understanding best when writing demands are reduced and when the focus stays on the physics rather than the mechanics of handwriting.	Offer a digital copy of materials if helpful, allow typing or speech-to-text where appropriate, and accept point-form or brief oral explanation when that gives a clearer picture of his learning. Pre-labelled graph paper or a partially structured graph template may help him focus on slope, area, units, and graph relationships.
Anish	Anish is social, involved, and strong in math. He often works best when the lesson is visually clear, well sequenced, and easy to track from step to step.	Keep the task sequence visible, pair oral directions with written directions, and use a short checklist so the stages of the investigation stay clear. Front seating during direct instruction, anchor examples, and extra processing time can

Student	Strengths-based view	Helpful supports in this lesson
		help him organize and communicate his thinking more confidently.
Sandra	Sandra is capable, athletic, and confident using translation technology. She is likely to participate most fully when the language demands of the task are scaffolded without reducing the conceptual work.	Allow translation software, reduce unnecessary reading load where possible, provide a visible word bank, and use sentence starters for explanations and reflection. Supportive peer grouping and strong visual examples can help her show what she understands about the physics while she continues to build academic language.
Juan	Juan is hardworking, intuitive, organized, and a strong self-advocate. He is likely to participate best when the environment is accessible and physical barriers are anticipated ahead of time.	Place materials within easy reach, make sure a normal-height chair or accessible workspace is available, and avoid assigning heavy lifting or two-handed setup tasks. He can contribute strongly in roles such as time caller, data checker, discussion lead, or graph interpreter.

Assessment reminder

When recording evidence, prioritize what the student understands about the motion graphs. Use the booklet, poster, oral explanation, observation, and reflection together so that one format does not become the only way a learner can demonstrate understanding.

Supply Teacher Guide: Motion Graphs Lab

Teacher-use setup and support guide for the original lesson materials.

This guide is designed to help a supply teacher run the lesson with confidence while keeping the focus on student strengths, participation, and the core physics learning.

Lesson focus

Students are investigating how position-time (p-t), velocity-time (v-t), and acceleration-time (a-t) graphs connect in one-dimensional motion.

- Slope of a p-t graph = velocity
- Slope of a v-t graph = acceleration
- Area under a v-t graph = displacement

Learning goal

Students will read, draw, and connect p-t, v-t, and a-t graphs and apply slope and area to explain motion.

Before class starts

☐ Set up a floor number line from 0 m to 8 m for the Minds On.

☐ Prepare one visible timer.

☐ Prepare group materials for the lab: track, toy car, graph paper, ruler, booklet, writing tools, and device for filming.

☐ Prepare ramp materials for Part 2.

☐ Prepare chart paper and markers for the gallery walk.

☐ Prepare the original 3-2-1 handout or lined paper for reflection.

Suggested lesson flow

1. Minds On: Walk the Graph

Use one student volunteer as the walker. Students watch the motion and sketch a quick p-t graph for each round. Remind them that time is on the x-axis, position is on the y-axis, and right is the positive direction. A quick whisper-check before boards up can help students feel ready.

2. Mini-review

State the Big 3 clearly: slope of p-t is velocity, slope of v-t is acceleration, and area under v-t is displacement.

3. Investigation A: level track

Students collect position-time data for a toy car, graph it, find slope, and connect the result to v-t and a-t. Encourage a best-fit line rather than dot-to-dot when appropriate.

4. Investigation B: ramp then level

Students repeat the process with a ramp section, mark the transition from Region A to Region B, and connect the changing p-t shape to the matching v-t and a-t graphs.

5. Poster and gallery walk

Each group creates a poster with aligned p-t, v-t, and a-t graphs, labels the regions, and includes one accurate slope statement and one accurate area statement. Students visit two posters and leave one accurate connection and one question or next step.

6. Consolidation: 3-2-1

Students write three graph connections, two fixes or glows for peer work, and one reflection about their learning.

Students to support with intention

Please use a strengths-based approach. The students below are capable learners who may benefit from a few specific supports during this lesson.

Michael: Very social, friendly, organized, and a strong self-advocate. He may benefit from reduced handwriting demands, typing, point form, or brief oral check-ins. Focus on his understanding rather than handwriting quality.

Anish: Very social, involved, and strong in math. He may benefit from visual directions, a clearly visible step order, and a moment of extra processing time before answering.

Sandra: Capable with translation software and benefits from supportive peers. She may benefit from a visible word bank, sentence starters, reduced unnecessary text load, and strong visual examples.

Juan: Hardworking, intuitive, organized, and a strong self-advocate. He may benefit from materials within easy reach and from roles that do not require heavy lifting or two-handed setup tasks.

What to look for while circulating

☐ Axes and units are labelled correctly.
☐ Students are using a best-fit line where appropriate.
☐ Students can explain slope correctly.
☐ Students can connect p-t to v-t and v-t to a-t.
☐ Students interpret area under v-t as displacement.
☐ Students can explain their thinking clearly.
☐ Students are contributing to their group and responding to peer feedback.

Quick setup map

Lesson phase	Materials to double-check
Minds On	Floor number line, visible timer, mini whiteboards or scrap paper, writing tools.
Action Part 1	Track, toy car, graph paper, ruler, booklet, device for filming, stopwatch if needed.
Action Part 2	Ramp starter, same track materials, chart paper, markers, sticky notes.
Consolidation	Original 3-2-1 handout or lined paper, visible sentence starters if helpful.

If technology or timing becomes an issue

- If a device is not working, students can still use a stopwatch and careful observation to gather simpler data.
- If time runs short, prioritize Part 1 on the level track, one strong graph conversion example, and the final 3-2-1 reflection.
- Keep the emphasis on explaining slope, area, and graph relationships rather than on finishing every possible calculation.

Important reminder

Keep the lesson focused on explanation and reasoning, not only on getting numerical answers. Students should be using graph language such as slope, area, sign, units, displacement, velocity, acceleration, Region A, and Region B.

Name:
Date:
Group Members:

Investigation: Toy Car Motion Data

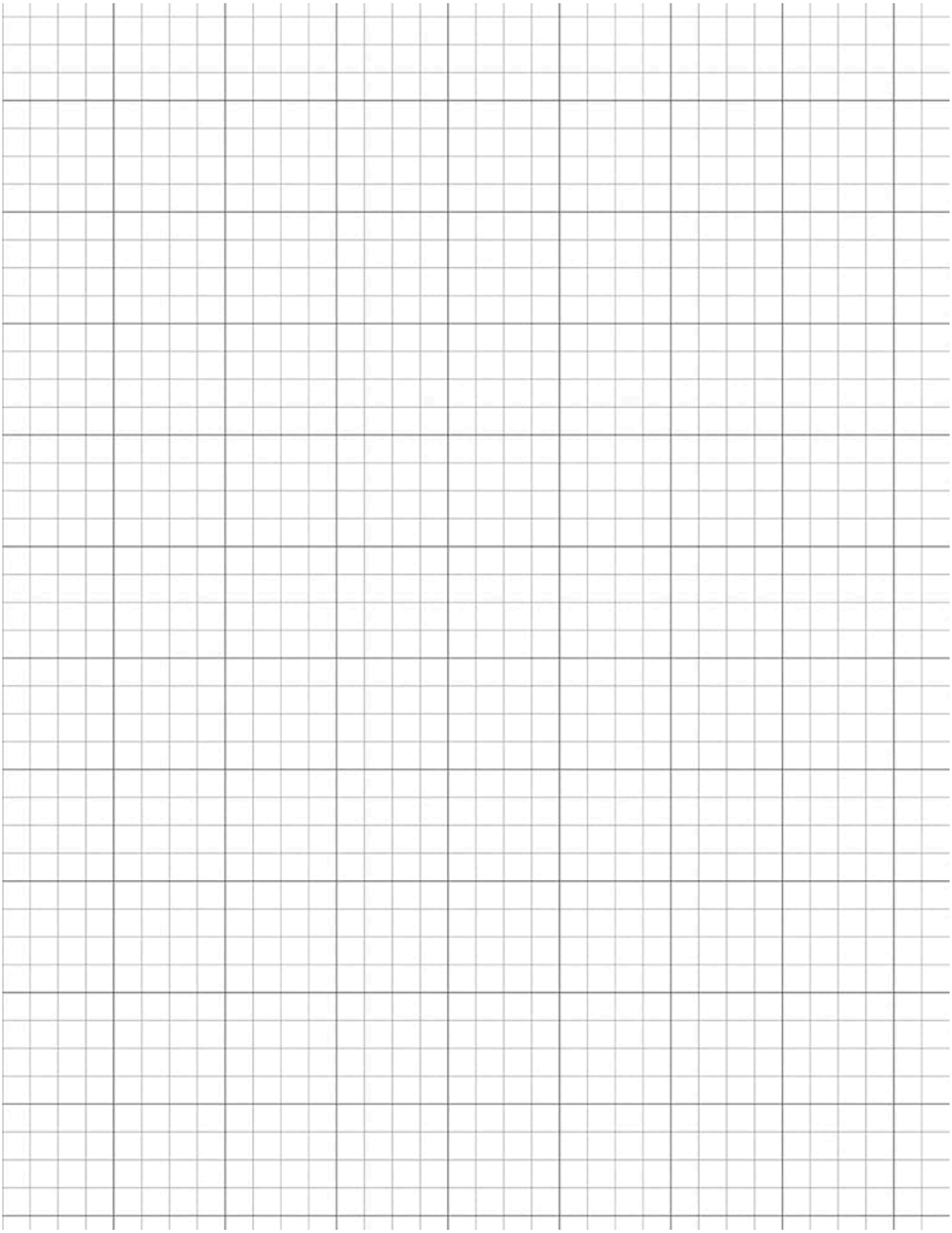
Objective: Collect position-time data for a toy car, graph it, and calculate its velocity

Roles	**Name of Group Member**
Timer (s)	
Spotter (s)	
Recorder (s)	

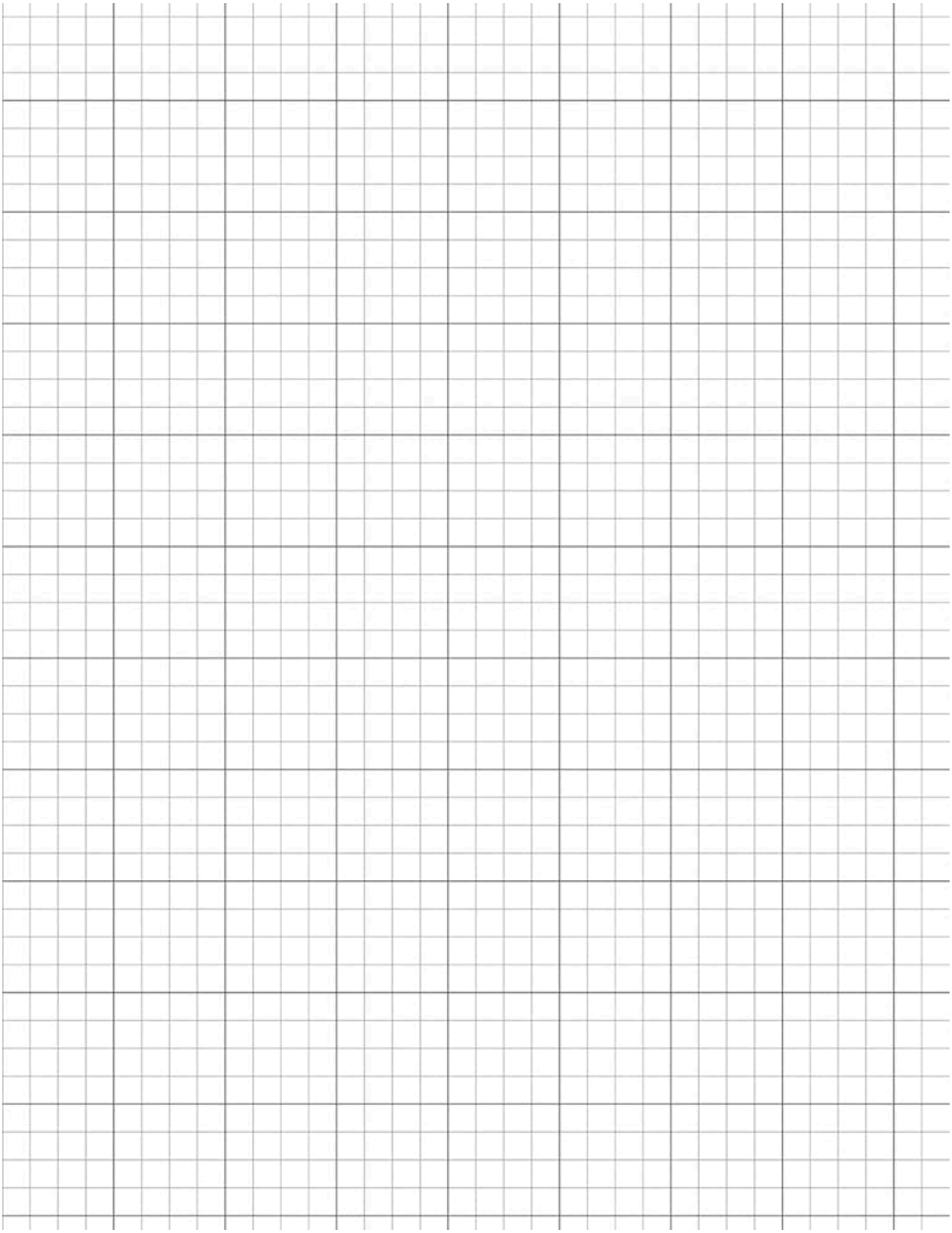
Trial 1 Data Table

Trial 2 Data Table

Trial 1 Graph (can replace with actual graph paper)



Trial 2 Graph



Questions
Point A: (position 1, time 1): **Point B**: (position 2, time 2): **Slope Calculation** = (position 1 - position 2) / (time 1 - time 2) = _____ **Units = _____**
1. **What does the slope represent in this activity? What would happen to the slope if the car went faster? Slower?**
2. **How did the data from Trial 1 and Trial 2 compare to one another? Why would we do multiple trials of the same process?**
3. **What does the area represent in this activity?**
4. **What is one potential source of error?**

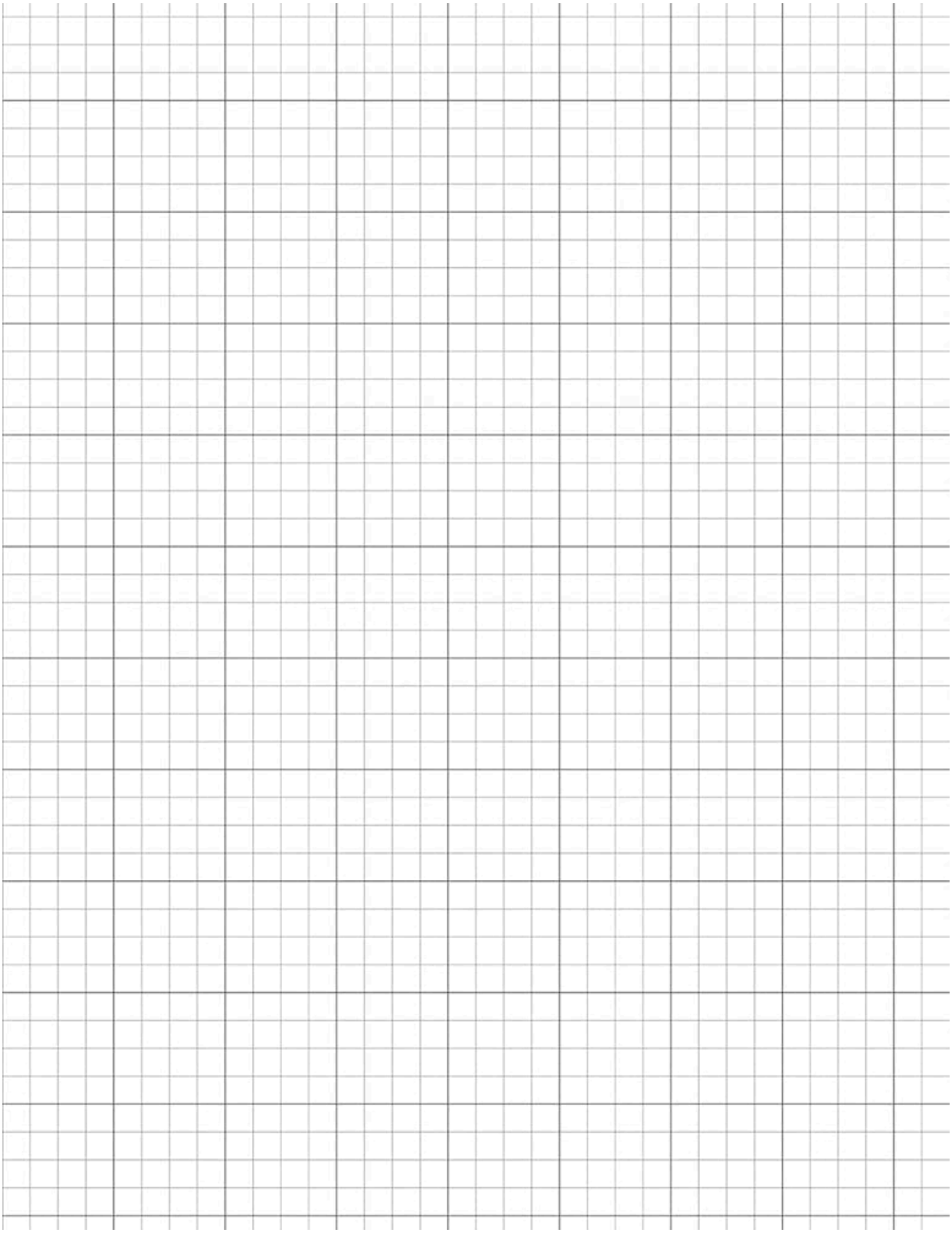
Objective: Collect position-time data for a toy car on a **ramp**, graph it, and calculate its velocity and acceleration

Roles	**Name of Group Member**
Timer (s)	
Spotter (s)	
Recorder (s)	

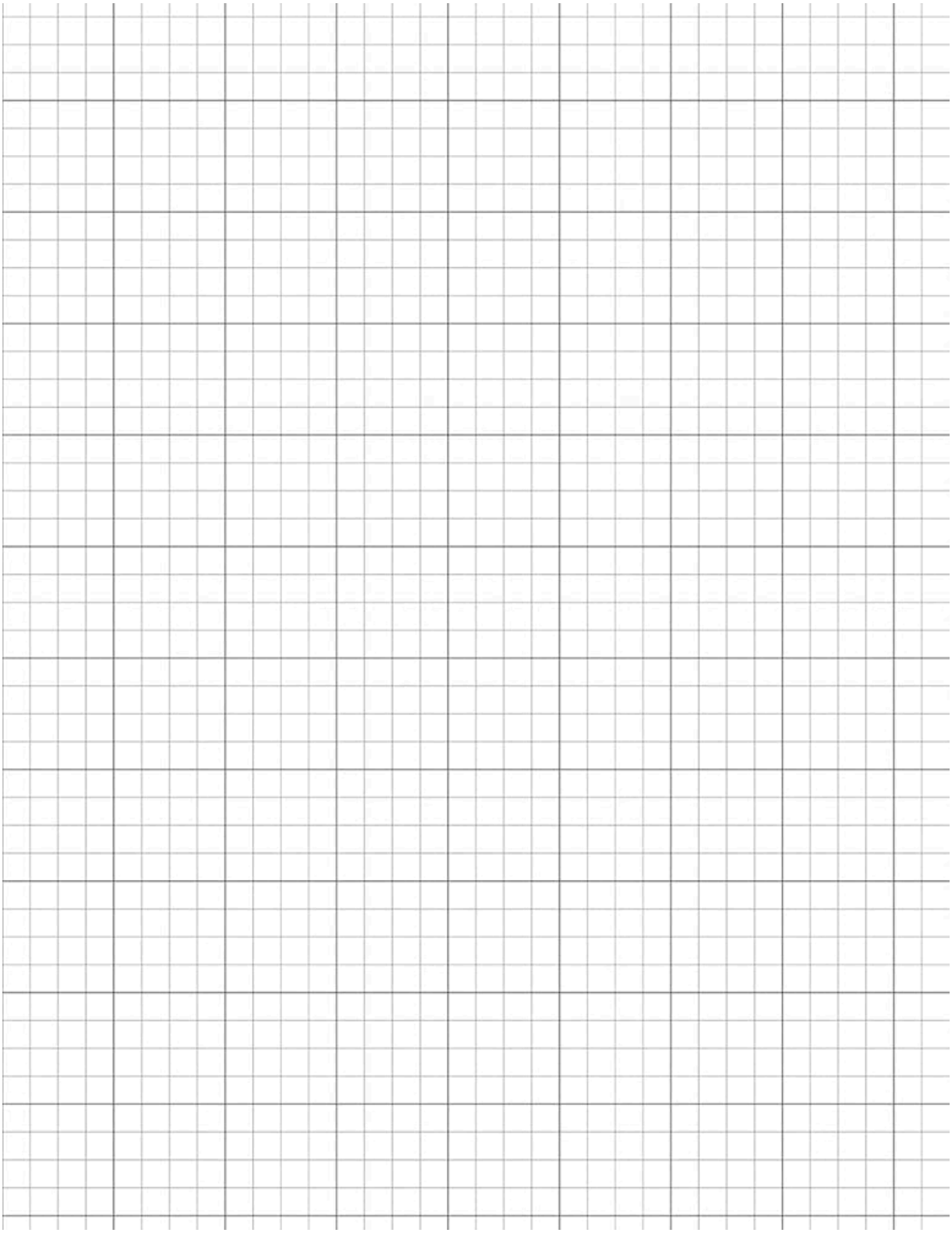
Trial 1 Data Table (Ramp)

Trial 2 Data Table (Ramp)

Trial 1 Graph (Ramp)



Trial 2 Graph (Ramp)



Questions (Ramp)

1. **Mark the transition.**

Ramp ends at position x = ___ m and time t = ____ s. Label this on both graphs.

2. **Level velocity from p-t (Region B).**

Pick two points in Region B and calculate the slope (with proper units and signs):

$$v_{\text{level}} = \frac{\Delta x}{\Delta t} = ____ \text{ m/s} \quad (\text{sign: } __)$$

3. **Shapes and zero-acceleration check.**

In one or two sentences, describe how p-t changes from Region A to B and explain what that requires from v-t and a-t in Region B.

5. **Ramp acceleration estimate (Region A)**

Use your v-t graph to estimate average acceleration during the ramp (or state its sign and why if you cannot get a number):

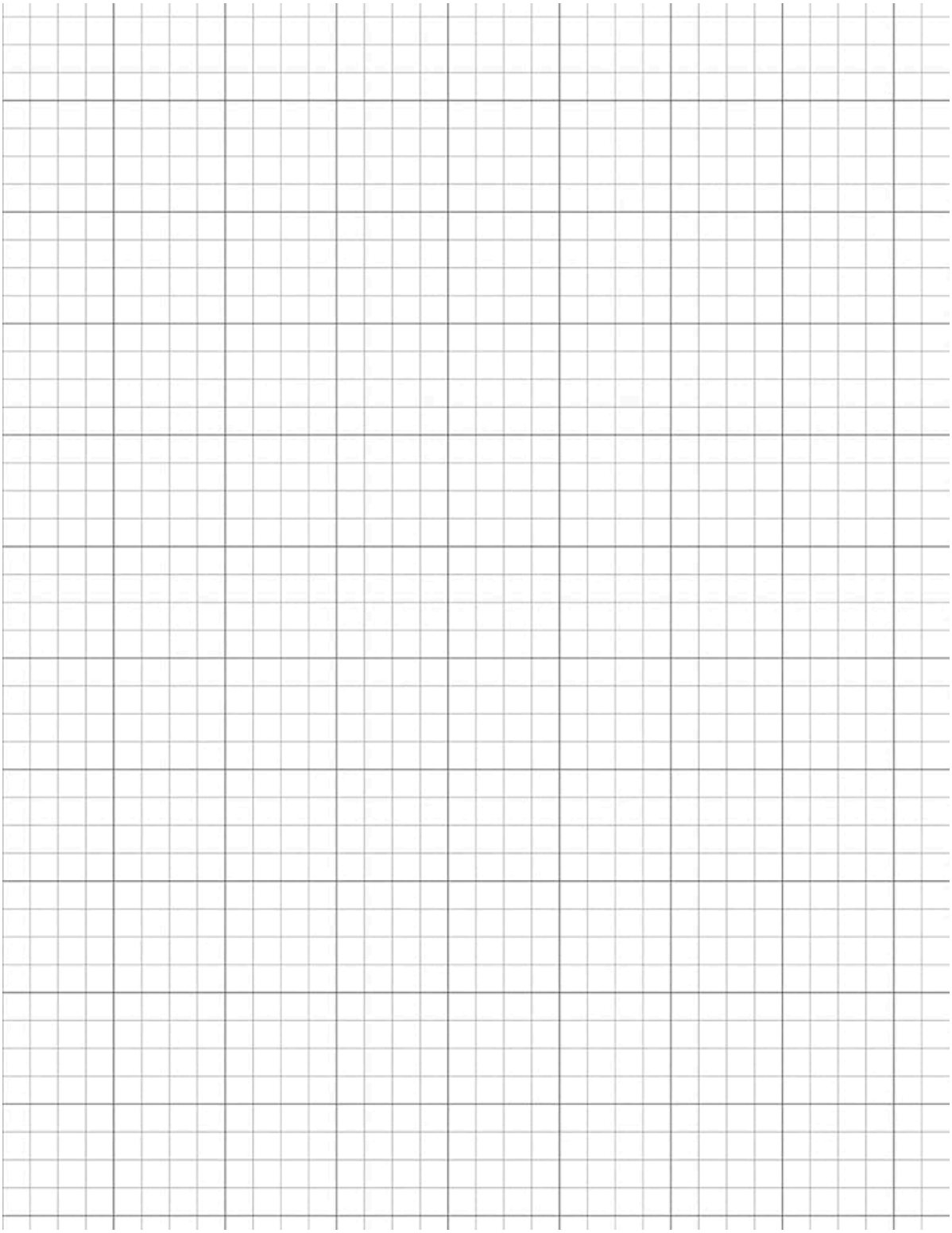
$$a_{\text{ramp}} \approx \frac{\Delta v}{\Delta t} = ____ \text{ m/s}^2 \quad (\text{sign: } __)$$

6. **Displacement by area (quick cross-check)**

For a chosen time interval [t1, t2]:
Area under v-t = ___m → predicted displacement
Change on p-t over the same times = __ m.
Matches? Yes / No

If **no**, one reason:

****Print additional graph paper if no graph paper is available in the classroom**



3-2-1 Connect and Reflect - Motion Graphs

Name: Date:

Keep your p-t, v-t, and a-t graphs visible. Write in full sentences using your own numbers, regions, and times.

A) 3 Connections (individual)

Write three sentences that connect your graphs. Sentence starters will be on the board (use them only if you find them helpful).

Connection 1
Connection 2
Connection 3

B) Peer Review - 2 Fixes or 'Glow + Next Step"

Trade papers. If you find errors or gaps, write TWO fixes. If you do not see errors, give a GLOW (strength) instead.

Reviewer name:

Fix 1 (or GLOW: something strong/effective)
Fix 2 (or GLOW: something strong/effective)

Reviewer check (tick any you checked): ☐ units ☐ signs ☐ axes labels ☐ slope vs height ☐ area on v–t ☐ region labels

C) **Metacognition (Pick two prompts to answer)**

Write in full sentences (on the back of the sheet)

1. One strategy that helped me today was…
2. One place I got stuck was… and I fixed it by…
3. I am still wondering / not sure why…

Self-assessment (circle one in each row)

Understanding of today's lab: 1 2 3 4 5

Motion Graphs Vocabulary Support Sheet

Keep this sheet beside your booklet during the lab. Use the words, symbols, and sentence starters when you explain your graphs.

Term	Student-friendly meaning	Quick reminder / symbol
position	Where the object is compared with the origin.	x or position (m)
displacement	How much the position changed, including direction.	Δx = final - initial
velocity	How fast position changes, including direction.	m/s; from p-t slope
acceleration	How velocity changes over time.	m/s^2; from v-t slope
slope	Rise over run; change in y divided by change in x.	Δy / Δx
area under v-t	Displacement over a chosen time interval.	rectangle or triangle
positive direction	The direction chosen as positive at the start.	keep signs consistent
negative direction	Motion opposite to the chosen positive direction.	below zero / decreasing
transition point	Where the motion changes from the ramp to the level track.	label on all graphs
best-fit line	A line that shows the overall trend, not every tiny wobble.	trend vs noise

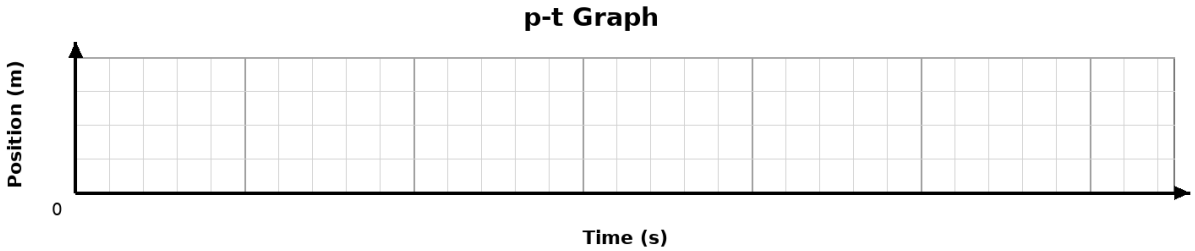
Helpful sentence starters

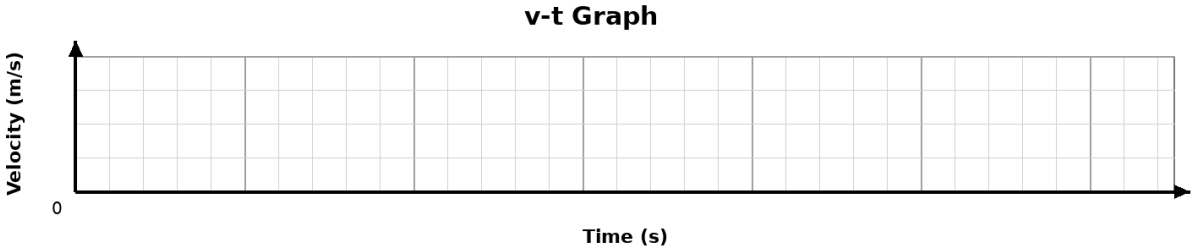
- The slope of my p-t graph shows ...
- The flat part of my v-t graph means ...
- The area under my v-t graph from t1 to t2 represents ...
- Region A shows ... because ...
- Region B shows ... because ...

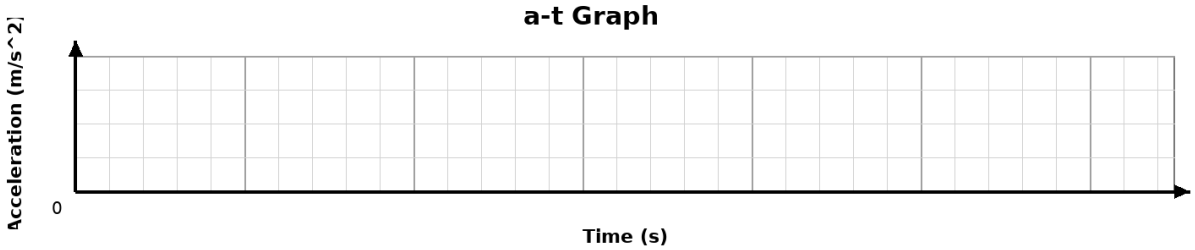
Scaffolded Graph Templates - Motion Graphs

Use these pages when students need axis labels, common graph titles, and aligned time axes already prepared.

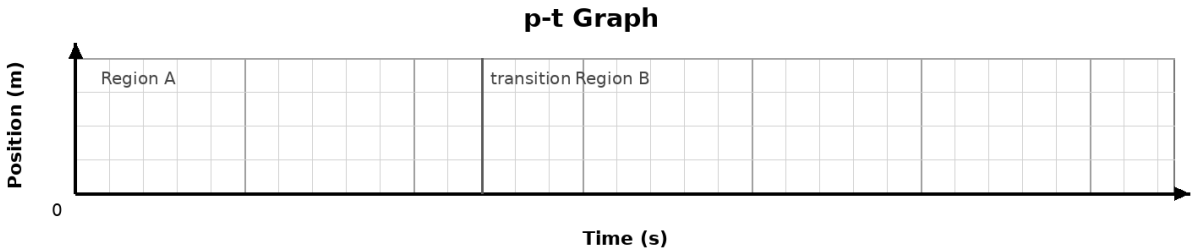
Template 1 - Standard p-t / v-t / a-t set

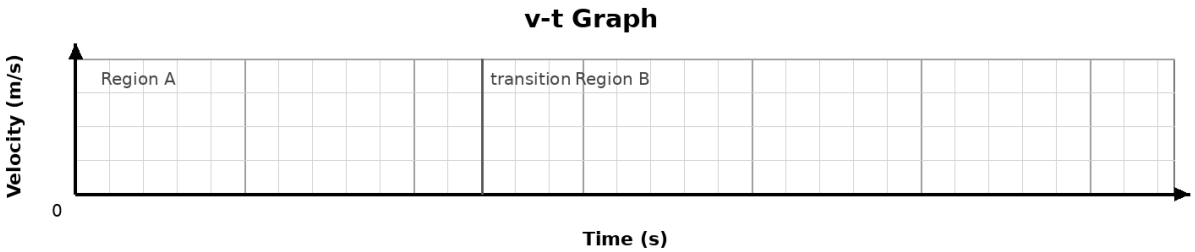


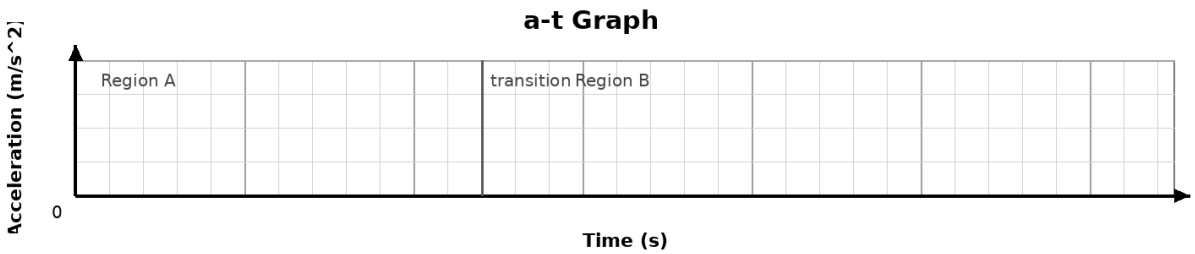




Template 2 - Ramp to level set with Region A / Region B support







Motion Groups - Assessment Checklist **Date:** **Teacher:**

A1 Axes/Units A2 Best-fit line A3 p–t→v–t (slope) A4 v–t→a–t (slope) A5 Area on v–t → Δx (A) Booklet (B) Poster (group)

Codes: 2 = secure 1 = developing 0 = with prompt N = not observed

Name	Group #	A1	A2	A3	A4	A5	Communication	Collaboration	(A) / 20	(B) / 12	Notes
1											
2											
3											
4											
5											
6											
7											
8											
9											
10											
11											
12											
13											
14											
15											
16											
17											
18											
19											
20											
21											

Anecdotal Notes - Conversations / Observation: Motion Stories

Use for quick quotes, misconceptions, next steps. Can use student number reference from the above page

(Print as needed)

Student #: Name: Evidence / quote: Next step / feedback given: Follow-up needed: Yes / No	Student #: Name: Evidence / quote: Next step / feedback given: Follow-up needed: Yes / No	Student #: Name: Evidence / quote: Next step / feedback given: Follow-up needed: Yes / No
Student #: Name: Evidence / quote: Next step / feedback given: Follow-up needed: Yes / No	Student #: Name: Evidence / quote: Next step / feedback given: Follow-up needed: Yes / No	Student #: Name: Evidence / quote: Next step / feedback given: Follow-up needed: Yes / No

Criteria for Student Booklet and Poster - Motion Graphs

A) Individual Lab Booklet (p–t → v–t → a–t) — Criteria

Weighting: /20 (Application 12, Communication 8)
Data & Tables (A) – complete, reliable, labeled (units, sign); two trials + clean data row. /5

5 Precise, complete; clear clean-data reasoning
4 Minor omissions; overall reliable
3 Some gaps/inconsistencies
1–2 Major missing data; labels/units absent

Graph Construction (A + C) – p–t with sensible scales, axes/units, best-fit (not dot-to-dot). /5

5 Accurate scales/labels; straight best-fit; neat & readable
4 Minor scale/label slip; trend still clear
3 Several errors; trend partly unclear
1–2 Misplaced axes/units; no best-fit

Conversions & Calculations (A) – p–t→v–t (slope value & sign), v–t→a–t (slope = 0 or ≠0, correct sign), v–t area → displacement and cross-check on p–t. /5

5 All conversions correct; area matches Δx with units
4 One small slip (rounding/sign), reasoning intact
3 Multiple slips; partial linkage
1–2 Little/no correct linkage

Explanations & Justifications (C) – short sentences that use slope/area/sign language to defend choices. /5

5 Clear, concise, correct physics language
4 Mostly clear; one term misused
3 Vague or mixed terms (height vs slope)
1–2 Minimal or off-target

Quick mark box (booklet): Data __/5 Graph __/5 Convert/Calc __/5 Explain __/5 → Total /20

B) Group Poster (three aligned panels) — Criteria

Weighting: /12 (Application 7, Communication 5)

Accuracy & Cross-Graph Alignment (A) – Part 1 level vs Part 2 ramp→level shown consistently (transition time marked). /5

5 Shapes & regions correct across p–t/v–t/a–t
4 One minor mismatch
3 Several mismatches; intent visible
1–2 Inconsistent/incorrect

Conventions & Clarity (C) – axes, units, time window; Region A/B labels; legible layout. /3

3 All conventions correct; easy to read
2 One convention missing
1 Two+ missing or hard to read

Justifications (C + A) – one slope statement (p–t) and one area statement (v–t) written on poster. /4

4 Both accurate with units/sign
3 One minor error
2 One missing or both vague
1 Both missing/incorrect

Quick mark box (poster): Align __/5 Clarity __/3 Justify __/4 → Total /12

****NOTE, For this mark all members of the group will be receiving the same mark unless mentioned otherwise in the notes.**